

Sara Vannah

sara.vannah [at] janusresearch.us

Education

Dartmouth College

PhD in Physics & Astronomy

Feb. 2023

Advisor: Marcelo Gleiser

Wellesley College

Bachelors of Arts in Astrophysics

May 2017

Advisor: Glenn Stark

Research Experience

Senior Research Associate

AER JANUS Research

Winter 2023 — Present

Developed patch for radio occultation processing tool to accept data from AWS Open Data registry. Compared refractivity retrievals between different processing centers. Performed air quality analysis projects for Texas Commission on Environmental Quality. Assessed the progression of the widths of the Tropics over decades through the locations of the Hadley cells in various models. Compared observations on models of Great Plains low-level jet to assess their ability to capture the physics of the jet.

Doctoral Research

Dartmouth College

Advisor: Marcelo Gleiser

Fall 2017 — Winter 2023

Developed information-theory based methods for analyzing cosmic microwave background and exoplanetary atmosphere spectra. Wrote information theory-based Markov Chain Monte Carlo for analyzing the cosmic microwave background. Mentored two undergraduate research projects

Student Researcher

Wellesley College

Advisor: Glenn Stark

Spring & Fall 2016

Demonstrated strong self-shielding effects in carbon monoxide. Performed reduction of sulfur dioxide spectroscopy data in search of a quantum-mechanically “forbidden” transition

KNAC Research Fellow

Vassar College through KNAC REU Program

Advisor: Colette Salyk

Summer 2016

Composed Python routines to perform spectral analysis on a series of protoplanetary disks. Used analysis to determine properties of the disks

Research Intern

Louisiana State University CCT REU Program

Advisor: Juhan Frank

Summer 2015

Developed an efficient, simplified analytic simulation of mass transfer in double white dwarf systems to supplement computationally expensive numerical simulations.

Publications

1. S.S. Leroy, S. Vannah, S. Ming, & P. Lin. (2026) "Diagnostics of the Great Plains low-level jet in atmospheric reanalyses", *in review in Journal of Applied Meteorology and Climatology*.
2. S.S. Leroy & S. Vannah. (2026) "Widening of the tropics in global surface-air winds", *Nature Scientific Reports* 16, 12344. [[doi:10.1038/s41598-026-43234-z](https://doi.org/10.1038/s41598-026-43234-z)]
3. S. Vannah, I. Stiehl, & M. Gleiser. (2025) "An Informational–Entropic Approach to Exoplanet Characterization", *Entropy* 27(4), 385. [[doi:10.3390/e27040385](https://doi.org/10.3390/e27040385)]

4. **S. Vannah**, S.S. Leroy, C. O Ao, E. R. Kursinski, K.J. Nelson, K.-N. Wang, & F. Xie. (2025) "The impact of differences in retrieval algorithms between processing centers on GNSS radio occultation refractivity retrievals in the planetary boundary layer", *Atmospheric Measurement Techniques* **18**, 4293–4310. [[doi:10.5194/amt-18-4293-2025](https://doi.org/10.5194/amt-18-4293-2025)]
5. **S. Vannah**, M. Gleiser, & L. Kaltenegger. (2024) "An information theory approach to identifying signs of life on transiting planets", *Monthly Notices of the Royal Astronomical Society: Letters* **528**(1), L4-L9. [[doi:10.1093/mnrasl/slad156](https://doi.org/10.1093/mnrasl/slad156)]
6. V. May, R. Sloboda, M. Tine, S. Streeter, D. Clemens-Sewall, **S. Vannah**, G. Goebel. (2022) "Feast or Famine Terrarium Project", Paper presented at 2022 ASEE Annual Conference & Exposition, Minneapolis, MN <https://peer.asee.org/40723>
7. M. Stephens, **S. Vannah**, & M. Gleiser. (2020) "Informational Approach to Cosmological Parameter Investigation" *Physical Review D* **102**, 123514 [[doi:10.1103/PhysRevD.102.123514](https://doi.org/10.1103/PhysRevD.102.123514)]
8. **S. Vannah** & C. Salyk. (2016) "A CO Spectral Analysis of Protoplanetary Disks", Keck Northeast Astronomy Consortium Symposium 2016, Wesleyan University, Middletown, CT. Paper published in conference proceedings

Posters & Presentations

*denotes invited talk

Information theory tools in the the search for life on exoplanets

Boston Area Planetary Sciences Meeting, Cambridge, MA (2025).

Quality Control in the Planetary Boundary Layer: An Intercomparison of GNSS Radio Occultation Retrievals Processed by UCAR, JPL, and ROM SAF

International Radio Occultation Working Group-10, Boulder, CO (2024) *talk* [slides](#)

An Information Theory Framework for Identifying Signs of Life on Transiting Exoplanets

Astrobiology Science Conference, Providence, RI (2024) *talk** [abstract](#)

Information theory as a tool in the search for life on exoplanets

MIT Planetary Lunch Seminars, MIT (2024) *talk*

Information Theory and the Search for Life in the Universe

Yuk Li Yung Lunch Seminar, CalTech (2024) *talk**

Comparing the Evolution of Earth-like Exoplanets Orbiting FGKM Stars: An Information Theory Approach

Planetary Science Informatics and Data Analytics, Villa de Cañada, Spain (2022) *talk* [slides](#), [recording](#)

A CO Spectral Analysis of Protoplanetary Disks

229th meeting of the American Astronomical Society, Grapevine, TX. (2017) *poster* [abstract](#)

Ancestral Astronomy: A Spectral Analysis of Protoplanetary Disks

Tanner Conference, Wellesley College, Wellesley, MA. (2016) *talk*

A CO Spectral Analysis of Protoplanetary Disks

Keck Northeast Astronomy Consortium Symposium, Wesleyan University, Middletown, CT.
(2016) *talk*

Simplified Simulation of Mass Transfer in Double White Dwarf Systems

227th meeting of the American Astronomical Society, Kissimmee, FL. (2016) *poster*
[abstract](#)

Awards & Honors

JANUS Energy Award

AER JANUS Research 2026

International Travel Grant

American Astronomical Society 2022

Selamawit Tsehaye Teaching Award

Dartmouth College 2020

Nominated for Guarini Teaching Award

Dartmouth College 2020

Dartmouth DCAL Outstanding Graduate Student Teacher Award

Dartmouth College 2018

New Hampshire Space Grant

NASA 2017

Sigma Xi research honor society nominee

Wellesley College 2017

Teaching, Service, & Outreach

Teaching Assistant

Winter 2018 — Fall 2022

Dartmouth College

Hanover, NH

- Led labs for nearly a dozen physics and astronomy courses, primarily introductory mechanics and electromagnetism
- Won both student and faculty-nominated awards for teaching
- Designed and led recitations
- Graded assignments quickly and effectively

Graduate Science Partner

Sept, 2021 – May, 2022

Dartmouth Rural STEM Educator Partnership

Hanover, NH

- Developed publishable [lesson plans](#) for a hands-on science curriculum for middle school students using Arduinos and terraria
- Met with middle school teachers, used feedback to adjust lessons to better fit rural students' needs
- Filmed and edited video versions of lessons for low-literacy students
- Visited local middle schools to interact with students, help implement lessons

Graduate Mentor

Spring 2021

Dartmouth Many Mentors

Hanover, NH

- Served as science mentor for rural Nevada high school

Public observing volunteer

2015-2017

Wellesley College

Wellesley, MA

- Led hands-on lessons for young children using the college's meteorite collection

Student Physics Department Representative

Fall 2016-Spring 2017

Wellesley College

Wellesley, MA

- Served as the student representative in department meetings
- Successfully changed seminar times to allow students with afternoon work-study jobs and athletes to attend

Society of Physics Students Treasurer

Fall 2016-Spring 2017

Wellesley College

Wellesley, MA

- Managed the finances for the club
- Helped organize and lead activities including both public outreach and activities within the department to help create a sense of belonging in physics

Specialized Skills

Programming Languages: Python (advanced), Mathematica (intermediate), MATLAB (beginner), Java (beginner)

Computational skills: object-oriented programming, data structures, machine learning, algorithms, high-performance computing

Technical skills: statistics, signal processing

Professional training

Astrostatics bootcamp at Pennsylvania State University, 2021

Future Faculty teacher training, Dartmouth Center for the Advancement of Learning, 2020

Scientific Communication course, Dartmouth College, 2018